

AI-DRIVEN PERSONALIZED CANCER TREATMENT SUMMARY

PATIENT PROFILE

Name: Robert Miller

MRN: MR-2026-1102-M

Gender: male

DOB: 15/3/1965

Age: 61

DOD: 5/2/2026

DISEASE CHARACTERIZATION & PATHOLOGY

Primary Diagnosis: Brain

Histological Subtype:

Clinical Stage:

WHO Grade: III

GENOMIC & VCF SUMMARY

Total Variants: 3

Actionable: 3

VUS:

Gene	Mutation Status	Clinical Impact	AI Weight
IDH1	Mutated	Favorable prognosis	High
MGMT	Unmethylated	Resistant phenotype	High
IDH1	Mutated (48%)	Pathogenic (Clonal)	High
PTEN	Lost (8%)	VUS (Low Frequency)	High
TP53	Mutated (32%)	Pathogenic (Subclonal)	High

FUNCTIONAL & PERFORMANCE STATUS

KPS Score: 70

ECOG Score: 2

Functional Category: Assisted

SYMPTOMS & COMORBIDITIES

Presenting Symptoms:

Comorbidities:

VOLUMETRIC ANALYSIS

Metric	Value	Metric	Value
Total Tumor Volume	112.68 cm ³	Active Core Volume	90.14 cm ³
Edema Volume	16.9 cm ³	Necrotic Fraction	5.63 %
Tumor Location	Left Temporal/Parietal	Sphericity Index	

BIOMARKER ANALYSIS

Biomarker	Status	Clinical Meaning
IDH1	Mutated	Biologically less aggressive
MGMT	Unmethylated	Poorer response to Temozolomide

Resection Extent	Subtotal	Impacts residual disease risk
IDH1	Mutated	VCF detected significance: Pathogenic (Clonal)
TP53	Mutated	VCF detected significance: Pathogenic (Subclonal)

AI-RECOMMENDED TREATMENT PLAN

{ "radiation": "Radiation therapy", "surgery": "Maximal safe resection", "systemic": "Chemotherapy (e.g., Temozolomide)", "targeted": "Targeted therapy (e.g., IDH1 inhibitors) for IDH1-mutated tumors" }

Rationale:

- The patient has a localized brain tumor with actionable variants (IDH1, PTEN, TP53). The primary treatment plan is based on the standard of care protocol for brain glioma, which includes maximal safe resection, radiation therapy, and chemotherapy. Targeted therapy may be considered for IDH1-mutated tumors.

OUTCOME PROJECTIONS

Overall Survival (Median): 15.2 Months

Quality of Life Score: 76.8/100

Progression-Free Survival: 9.1 Months

Primary Toxicity Risk: Monitor for fatigue and nausea